

Trailer Tetris

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Now that the SQUIRM is back up and running, the field-deployment team is excited to get back out to sampling for helminths. They realized they need to make some space in the messy field-trailer to fit the newly repaired unit, and so several technicians have been assigned the job of cleaning of sections of the trailer. Every section of the trailer has a unique ID number, and each technician is assigned a range of section IDs.

However, as some of the technicians compare their section assignments with each other, they've noticed that many of the assignments overlap. To try to quickly find overlaps and reduce duplicated effort, the technician pair up and make a big list of the section assignments for each pair (your puzzle input).

For example, consider the following list of section assignment pairs:

2-4,6-8
2-3,4-5
5-7,7-9
2-8,3-7
6-6,4-6
2-6,4-8

For the first few pairs, this list means:

- Within the first pair of technicians, the first tech was assigned sections 2-4 (sections 2, 3, and 4), while the second tech was assigned sections 6-8 (sections 6, 7, 8).
- The technicians in the second pair were each assigned two sections.
- The technicians in the third pair were each assigned three sections: one got sections 5, 6, and 7, while the other also got 7, plus 8 and 9.

This example list uses single-digit section IDs to make it easier to draw; your actual list might contain larger numbers. Visually, these pairs of section assignments look like this:

.234. 2-4
.....678. 6-8

.23. 2-3
...45. 4-5

...567.. 5-7
.....789 7-9

.2345678. 2-8
..34567.. 3-7

.....6... 6-6
...456... 4-6

.23456... 2-6
...45678. 4-8

Some of the pairs have noticed that one of their assignments fully contains the other. For example, 2-8 fully contains 3-7, and 6-6 is fully contained by 4-6. In pairs where one assignment fully contains the other, one technician in the pair would be exclusively cleaning sections their partner will already be cleaning, so these seem like the most in need of reconsideration. In this example, there are 2 such pairs.

Below your actual input is read in as a list. Each element is a vector of 4 numbers; the first two define the beginning and ending sections of first member of the pair's assignment. The second two define the same for the second member of the pair.

In how many assignment pairs does one range fully contain the other?

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load(file="input.RDA")
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Part II: To get an idea of how much duplicate work is planned, the technicians would like to know how many assignment don't overlap at all.